Tridib Banik

+1 647-765-2966 | banikt@mcmaster.ca | linkedin.com/in/tridib-banik17 | github.com/tridibbanik17

EDUCATION

McMaster University *Sept. 2023 – Apr. 2028*

Bachelor of Engineering, Software Engineering Co-op (CGPA: **3.9 / 4.0**) Hamilton, ON

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Object-Oriented Programming, Requirements Engineering & Security, Systems Design, Concurrent Systems, Software Testing, Controls.

EXPERIENCE

Research Assistant Intern *June 2025 – Aug. 2025*

CMHT Lab, McMaster Automotive Resource Centre Hamilton, ON

- Co-developed a 2024 Ram ProMaster EV Battery Management System (BMS) manuscript, concluding that core logic uses **0.99% flash** and **9.23% RAM**, by benchmarking **7 subsystems** on an NXP automotive microprocessor.
- Designed a **Simulink** model of our team's custom **105s34p** battery pack, enabling safe and realistic verification of the complete BMS logic prior to physical deployment on the Ram ProMaster electric van.
- Accelerated onboarding for **5+** new team members by authoring **11** technical presentations totaling **308 slides** detailing software installation and hardware setup for a custom BMS.

Software Engineer Intern *Sept. 2024 – Apr. 2026*

Battery Workforce EV Challenge (McMaster Team) Hamilton, ON

- Validated **4** model-based **Simulink** subsystems, including Contactor Control, Cell Balancing, Fast Charging, and Fault Diagnosis, against competition specs, achieving a **95%+** unit test pass rate across **100%** of requirements.
- Shifted team testing from manual execution to automated continuous integration by architecting a **5-stage GitLab CI/CD pipeline** (verify, build, test, package, deploy) hosted on a dedicated **GitLab Runner**.
- Secured a **$6,500 merit scholarship** for technical contributions in subsystem design, system integration, and Model-in-the-Loop (MiL) testing, supporting the team's **2nd place** finish out of **12 North American universities**.

Teaching Assistant *Jan. 2026 – Apr. 2026*

Integrated Cornerstone Design Projects course, McMaster University Hamilton, ON

- Mentored **23 students** across **5 teams** through a **6-milestone** wastewater filtration project, guiding designs from problem definition to final material selection using objective trees, regulatory compliance, and decision matrices.

PROJECTS

TrafficLightRL | Python, OpenAI Gymnasium, Stable-Baselines3, TensorFlow

- Architected a modular **Reinforcement Learning** pipeline with isolated environment, training, and evaluation components, presenting the framework to **320+ attendees** at the 2025 Canadian Undergraduate Conference on AI.
- Optimized urban traffic flow against a fixed-cycle baseline by training a **PPO** agent, reducing vehicle wait times by up to **29.55%** and cutting CO_2 emissions by **35.56%** across **3** distinct traffic densities.

CNC Data Pipeline | Python, Flask, React, PostgreSQL, Docker, Scikit-learn

- Engineered an **Industry 4.0** analytics platform that ingested simulated Fanuc, Haas, and Mitsubishi CNC telemetry into **PostgreSQL** via **Flask REST APIs**, tracking real-time OEE metrics, i.e., spindle load and tool wear.
- Developed a secure **React** dashboard (Recharts) with **JWT** role-based access control, integrating **Scikit-learn IsolationForest** models to execute real-time anomaly detection for predictive machine maintenance.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, MATLAB, HTML, CSS

Frameworks: React.js, Node.js, Flask, Django, TensorFlow, Scikit-learn, NumPy

Tools: Simulink, Simscape, Docker, Git, PostgreSQL, Linux, VS Code, Cursor, Kiro, Power Automate